Project Title

Plagiarism Detection System using Rolling Hash Techniques

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Submitted by
Prakshitha Malla(2520030342)
Datla Sri Sai Purna Mahija(2520030261)
Kheerthi Ganta(2520030383)

Team Number: 10
Section: 5

Under the guidance of

D Swathi Mam

Assistant Professor, CSE



ABSTRACT

Academic and intellectual integrity relies on the effective identification of copied content across large document collections. This project presents a robust plagiarism detection system developed specifically to identify duplicate passages using efficient rolling hash techniques. Grounded in the principles of the Rabin-Karp string-matching algorithm, the application generates polynomial hash values for sliding windows of text, enabling rapid identification of matching segments across multiple documents. To ensure maximum accuracy, the system implements rigorous collision handling mechanisms, such as double hashing, preventing false positives when different text passages produce identical hash values. Adhering strictly to core system constraints, all underlying hash structures, string processing algorithms, and array manipulations are engineered entirely from scratch without relying on built-in standard library utilities like `java.util.*`. Beyond detecting matches, the engine calculates comprehensive similarity percentages to quantify document overlap accurately. Furthermore, it visually highlights duplicated passages within the interface to provide intuitive, actionable output for end users. Ultimately, this project demonstrates a high-performance, sub-quadratic approach to multi-document comparison and practical string analytics.

Signature of the Guide